

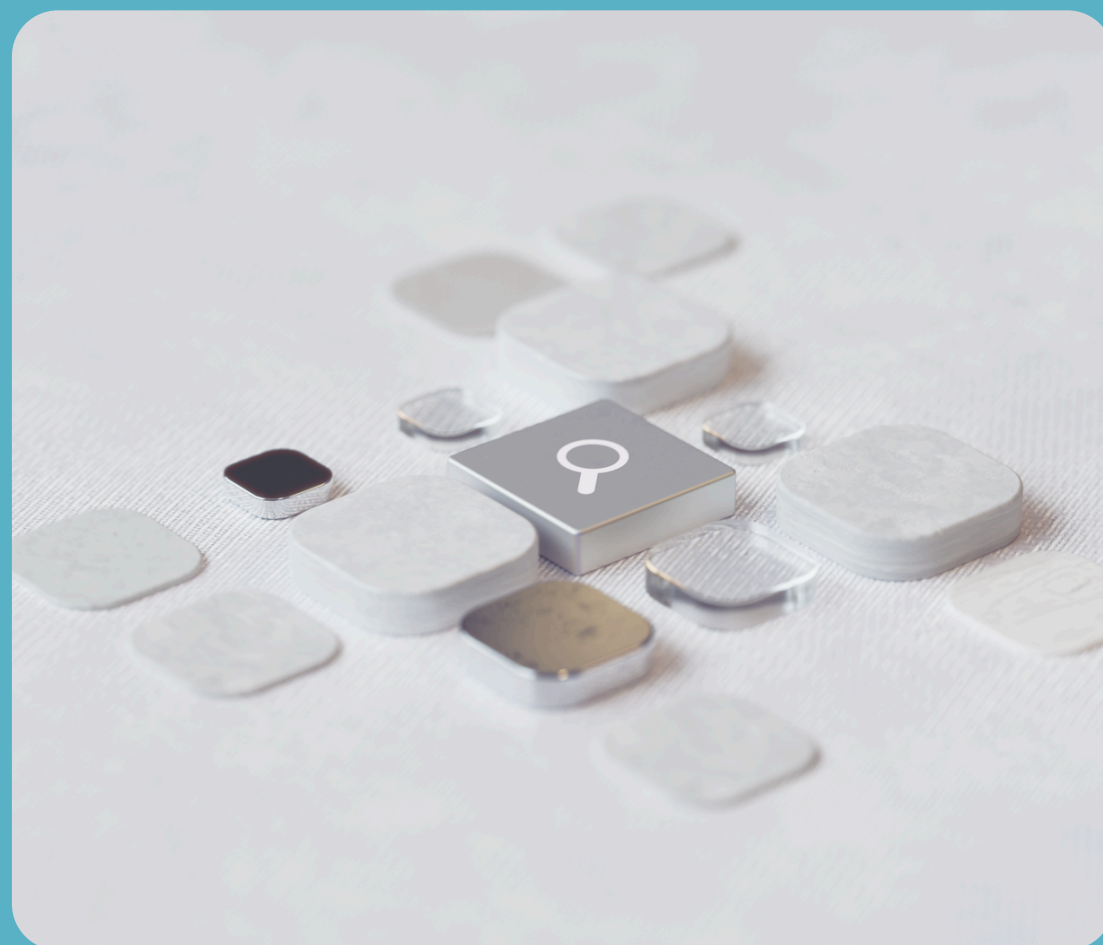
Plant Disease Classification Using Deep Learning



Problem Statement & Objectives

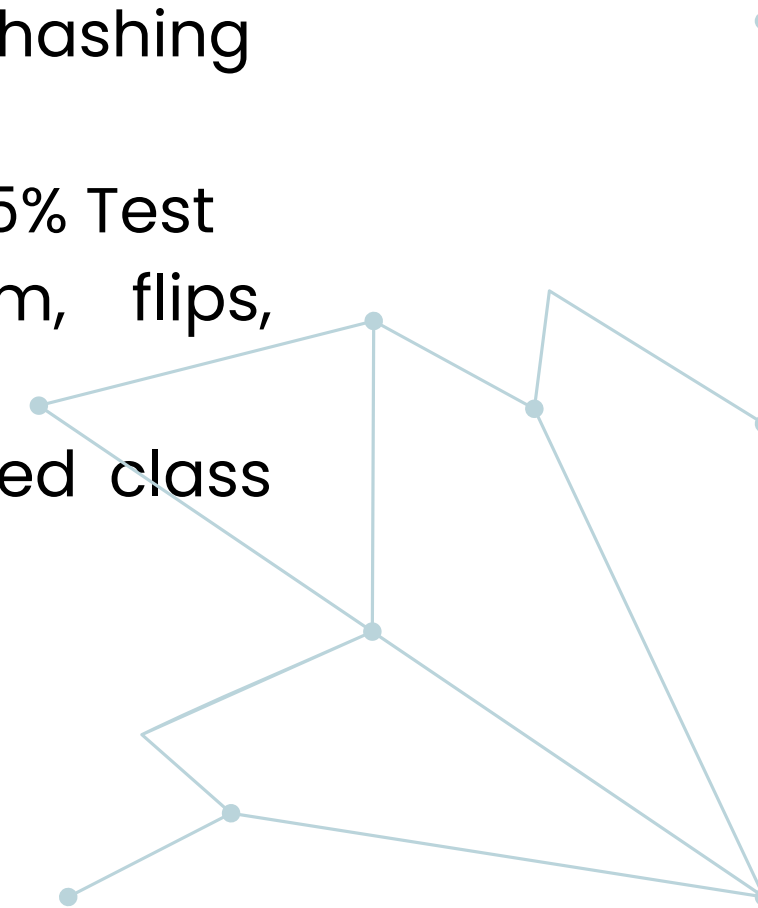
- Problem: Early detection of plant diseases is crucial for agricultural productivity
- Objectives:
 - Build and compare multiple deep learning models for plant disease classification
 - Evaluate performance across 38 disease classes
 - Implement data augmentation and class balancing techniques
- Dataset: PlantVillage with color images of healthy and diseased plant leaves





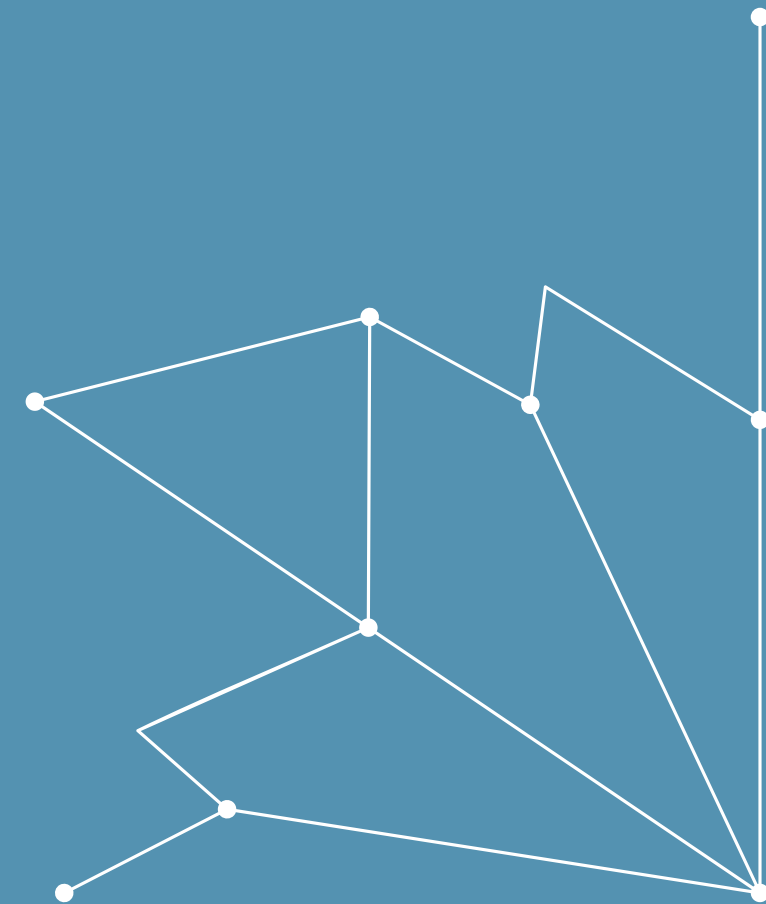
Dataset Overview

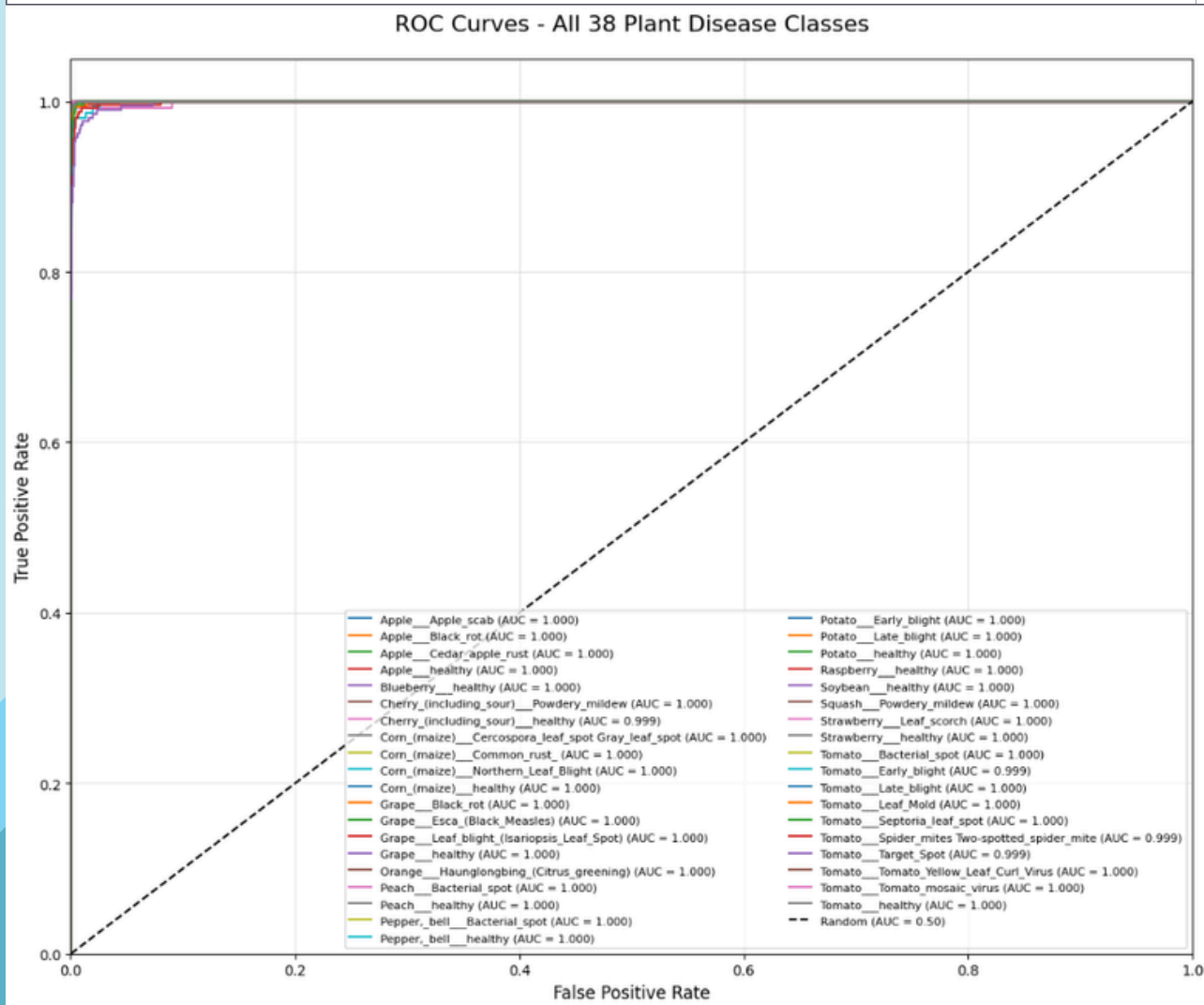
- Source: PlantVillage Dataset (Kaggle)
- Classes: 38 plant disease categories
- Preprocessing:
 - Removed duplicate images using MD5 hashing
 - Image size: 224×224 pixels
 - Data split: 70% Train / 15% Validation / 15% Test
 - Augmentation: Rotation, shifts, zoom, flips, brightness variation
- Class Imbalance: Handled using computed class weights



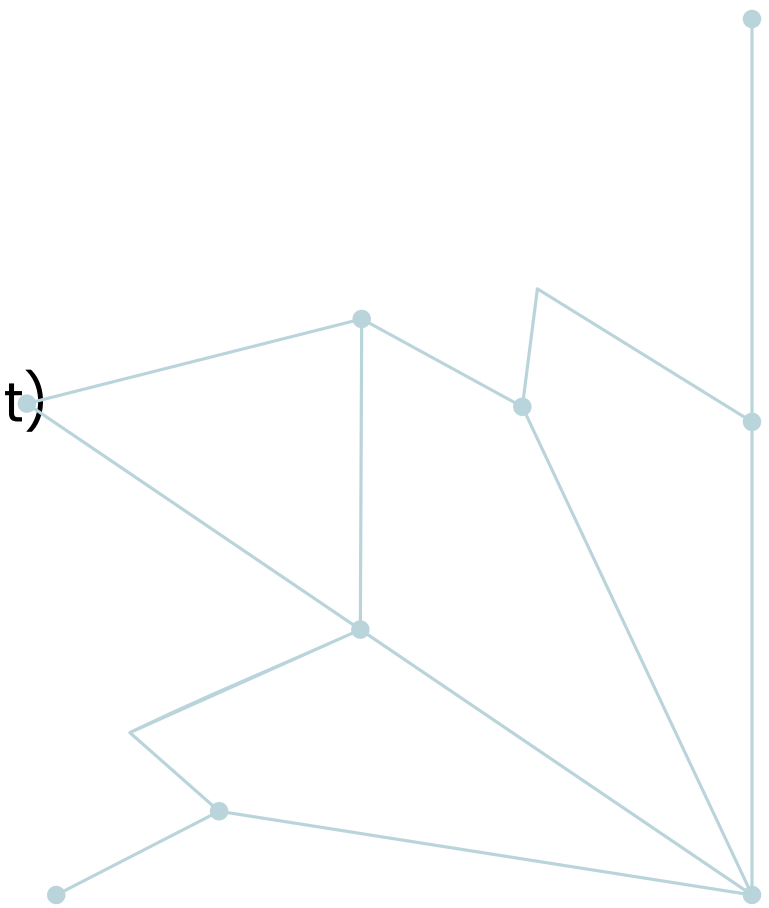
Models Implemented

- VGG-19 (Built from scratch)
 - 19 layers (16 conv + 3 dense)
 - Classic CNN architecture
- ResNet-50 (Transfer Learning + Fine-tuning)
 - Pre-trained on ImageNet
 - Two-phase training approach
- MobileNetV2 (Transfer Learning)
 - Lightweight architecture
 - Efficient for deployment
- Vision Transformer (ViT-B16) (Transfer Learning)
 - Attention-based architecture
 - Modern transformer approach
- Inception V1 (GoogLeNet) (Transfer Learning)
 - Multi-scale feature extraction



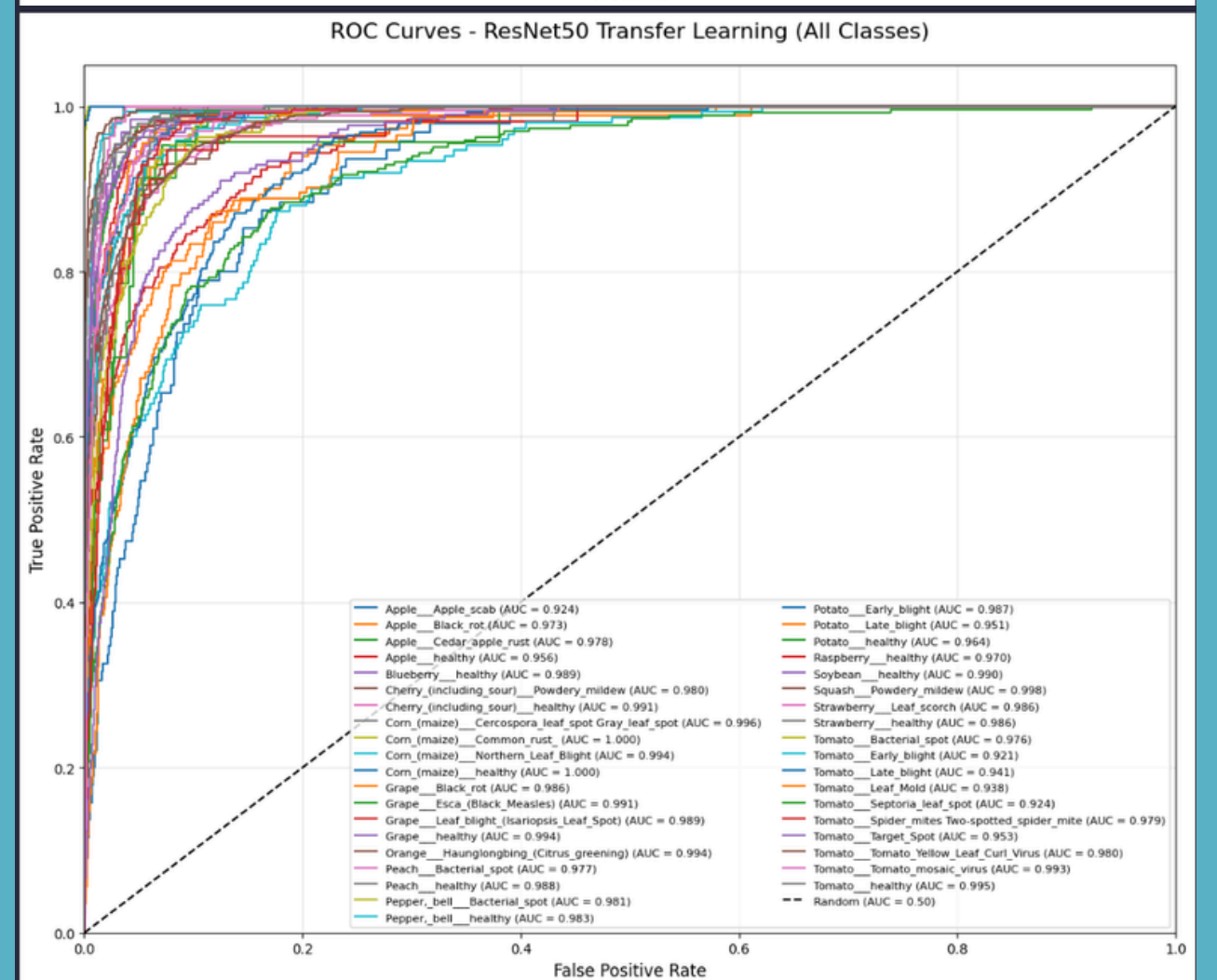
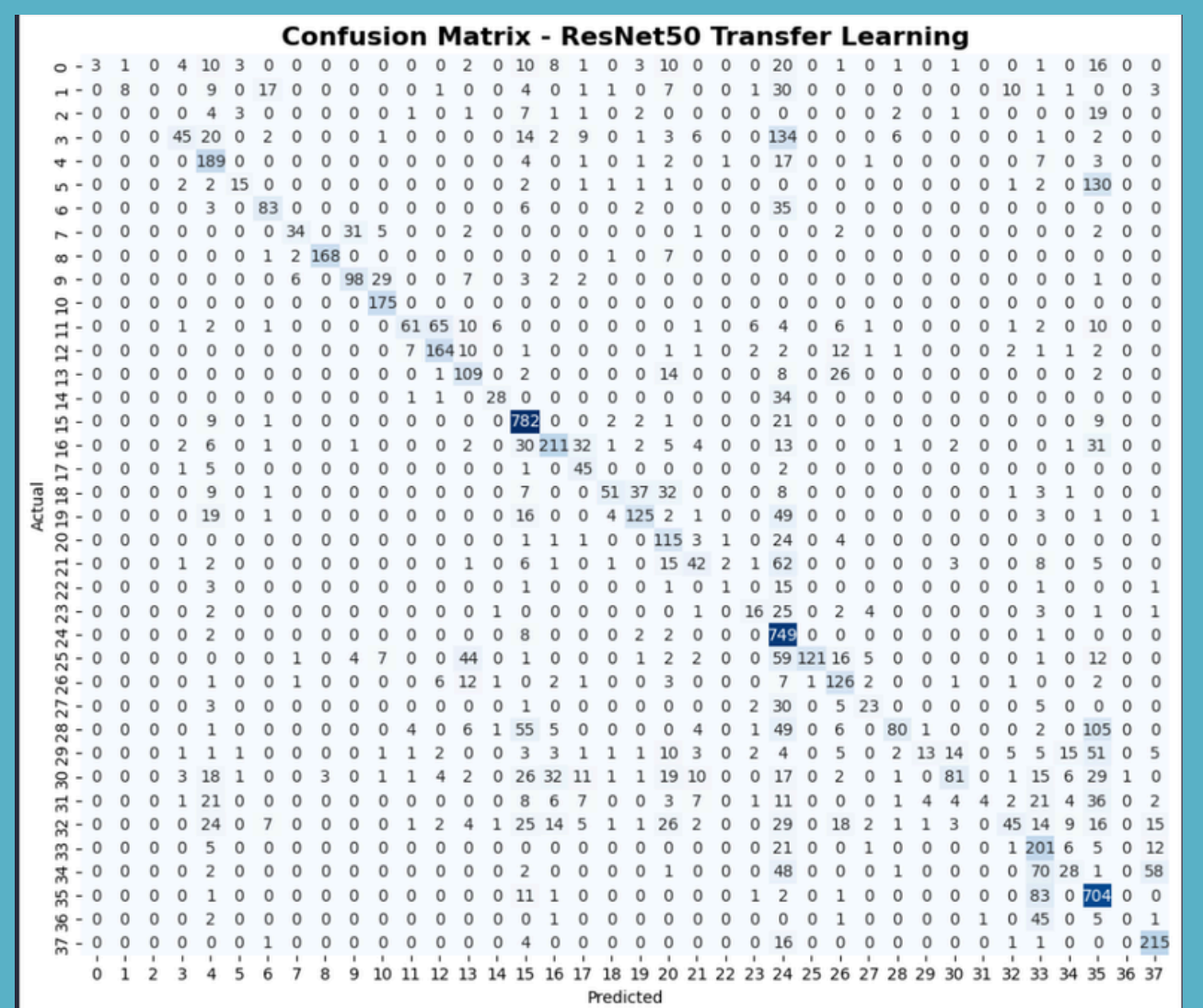
[illegible]

- Architecture: Built from scratch with 5 convolutional blocks
- Training: 30 epochs with early stopping
- Results: Best
 - Precision : **97%**
 - Recall : **97%**
 - F1-Score : **97%**
- Observations:
 - Good baseline performance
 - High computational cost
 - Prone to overfitting (mitigated with dropout)



ResNet-50 Results

- Approach: Two-phase training
 - Phase 1: Transfer learning (frozen base)
 - Phase 2: Fine-tuning (last 50 layers unfrozen)
- Results:
 - Precision : **62%**
 - Recall : **47%**
 - F1-Score : **46%**
- Observations:
 - Best overall performance
 - Effective feature extraction with residual connections
 - Fine-tuning improved accuracy by ~9%



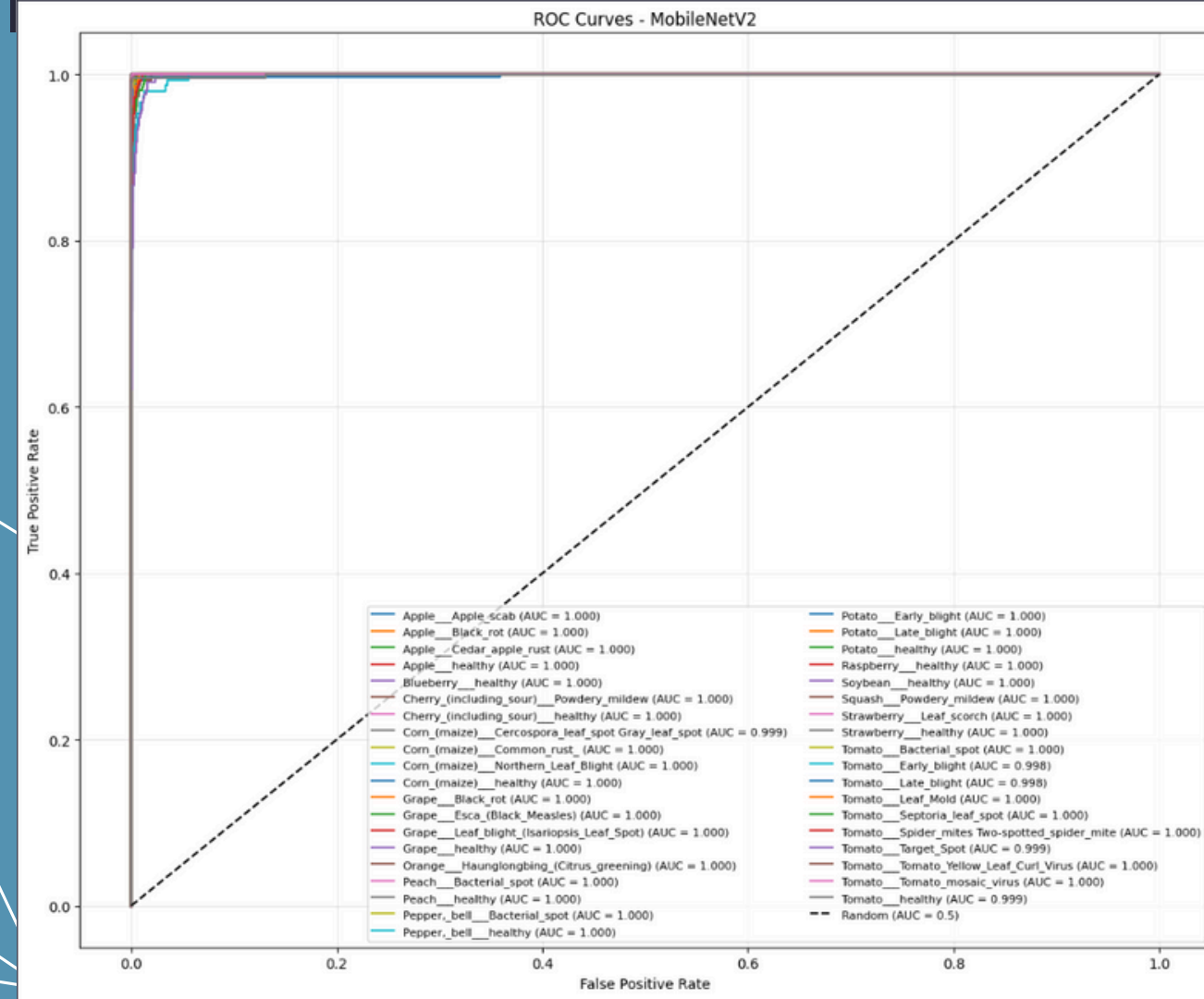
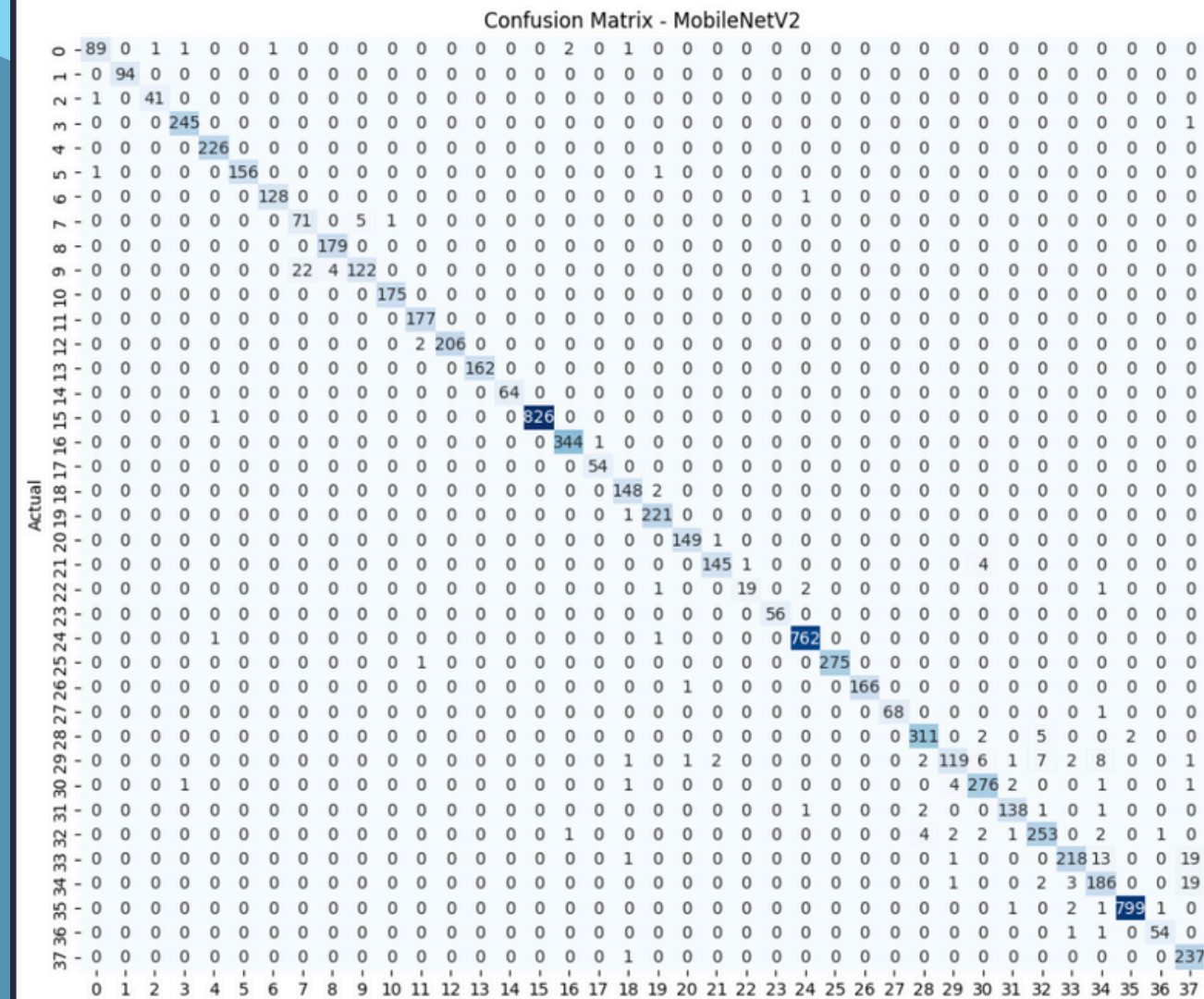
MobileNetV2 & ViT

MobileNetV2:

- Frozen training: 20 epochs
- Fine-tuning: 15 epochs (last 40 layers)
- Results: Best
 - **Precision : 97%**
 - **Recall : 97%**
 - **F1-Score : 97%**
- Lightweight and efficient for mobile deployment

Vision Transformer (ViT-B16):

- Frozen training: 10 epochs
- Fine-tuning: 10 epochs (last 4 layers)
- Results: Best
 - **Precision : 98%**
 - **Recall : 98%**
 - **F1-Score : 98%**
- Attention mechanisms capture global patterns
- Competitive with CNNs



MobileNetV2 & ViT

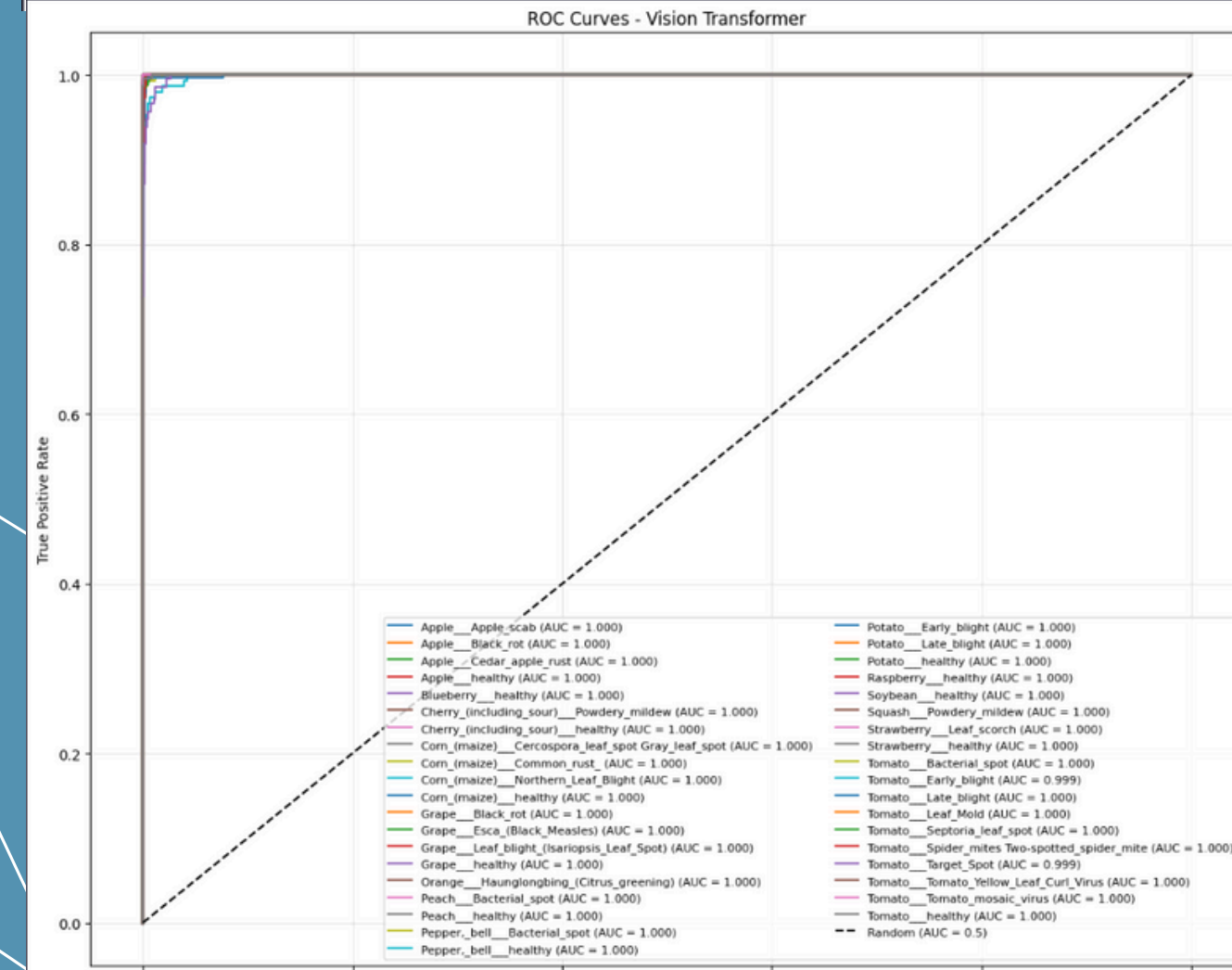
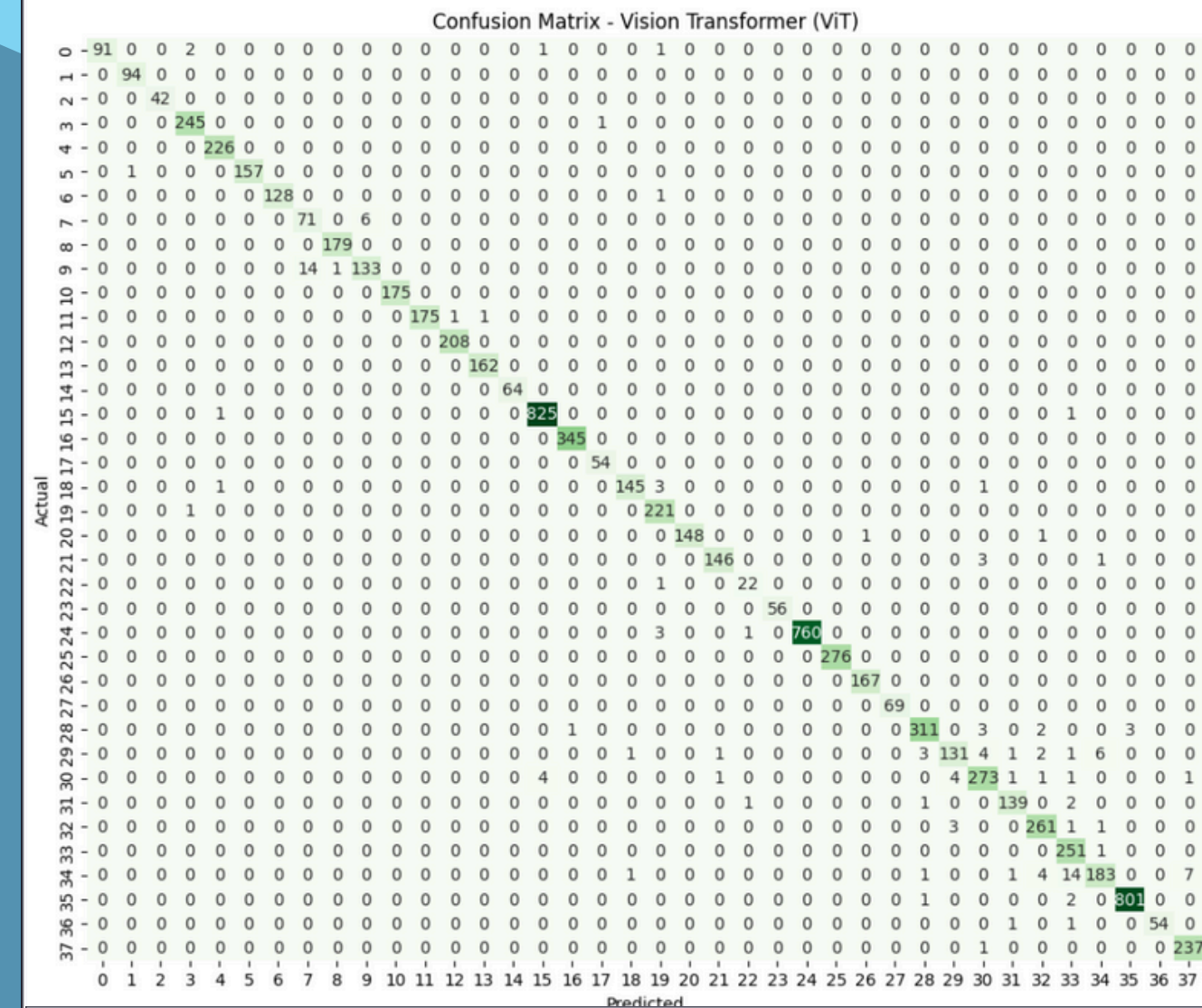
Results

MobileNetV2:

- Frozen training: 20 epochs
- Fine-tuning: 15 epochs (last 40 layers)
- Results: Best
 - **Precision : 97%**
 - **Recall : 97%**
 - **F1-Score : 97%**
- Lightweight and efficient for mobile deployment

Vision Transformer (ViT-B16):

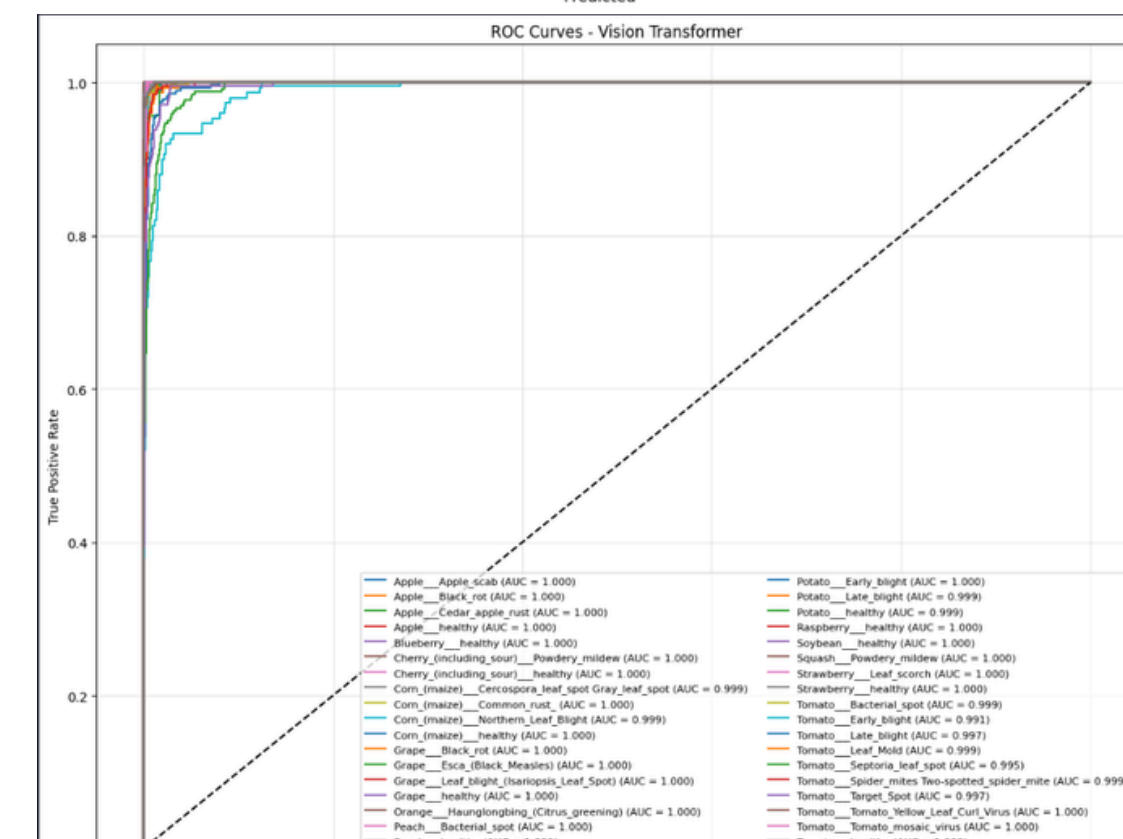
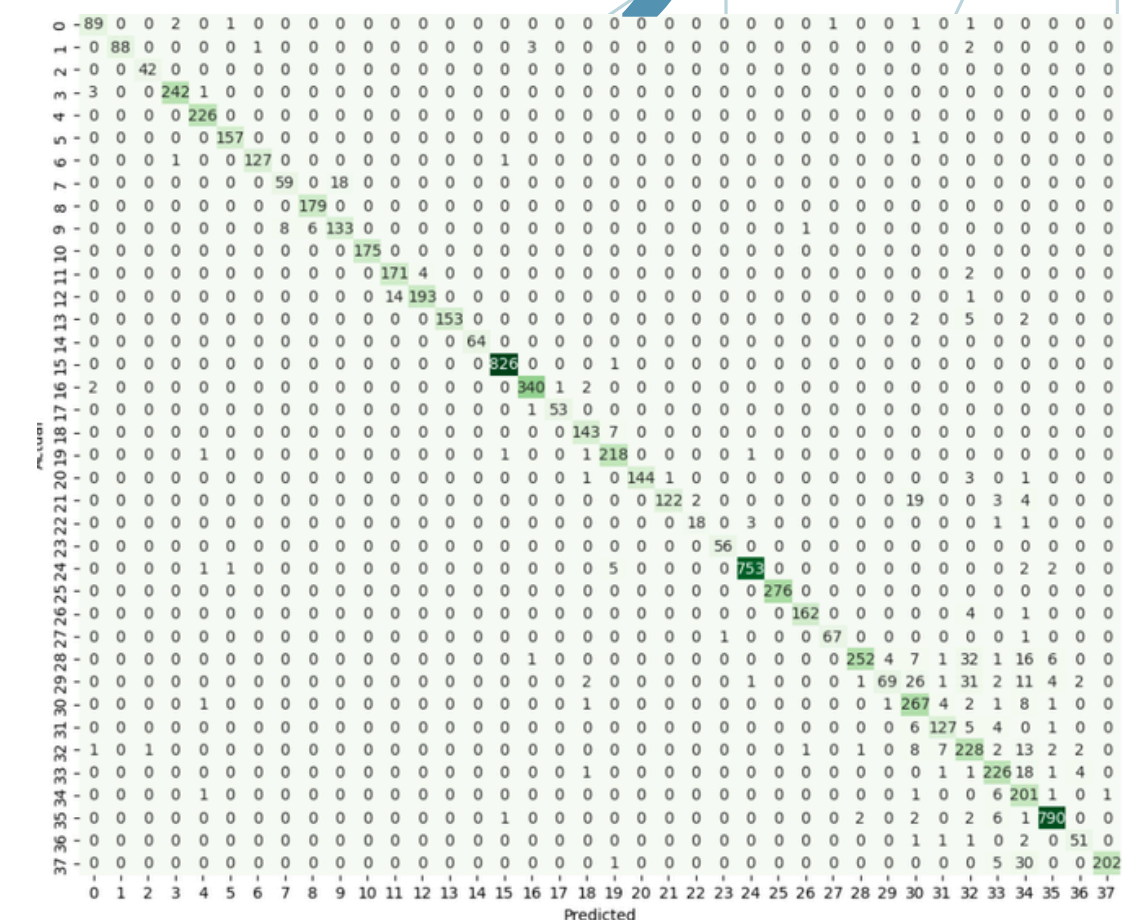
- Frozen training: 10 epochs
- Fine-tuning: 10 epochs (last 4 layers)
- Results: Best
 - **Precision : 98%**
 - **Recall : 98%**
 - **F1-Score : 98%**
- Attention mechanisms capture global patterns
- Competitive with CNNs



Inception V1 (GoogLeNet)

Results

- Architecture: Multi-scale convolutions with TensorFlow Hub
- Training: 7 epochs with base frozen
- Results: Best
 - **Precision : 94%**
 - **Recall : 93%**
 - **F1-Score : 93%**
- Observations:
 - Efficient multi-scale feature extraction
 - Good performance with fewer epochs
 - Inception modules enable parallel processing



Conclusions & Future Work

Conclusions:

- Data augmentation and class balancing improved generalization
- Model choice depends on deployment constraints (accuracy vs. efficiency)

Future Work:

- Deploy best model as mobile/web application
- Test on real-world field images
- Implement ensemble methods
- Add explainability (Grad-CAM) for interpretability



The Journey Begins With a Question

Thank you for exploring the world of AI. We've only scratched the surface—now let's dive deeper through your questions, reflections, and discussions. AI isn't just the future—it's the now.

